Plastic recurrent excitation fails the local SHD gate

exp073 · 20 July 2026 · Draft

Abstract

This exploratory experiment tested the next biologically plausible path after exp071/exp072: keep the cumulative-potential readout, keep the historical one-seed validation-only SHD split, and make recurrent E→E excitation both nonzero and plastic while preserving Dale signs. The required local smoke stage killed the experiment before RunPod use. PING remained finite and active, but the matched COBA cell repeatedly produced skipped/non-finite updates even after reducing the requested W__EE scale from `0.3 0.01` through `0.03 0.01`, `0.003 0.001`, and finally `0.0003 0.0001`. No paid compute was run. The result is therefore a design-gate failure, not evidence about eight-epoch or forty-epoch SHD accuracy.

Goal prompt

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/goal Design and execute a new exploratory SHD experiment on a new branch/PR,
testing whether nonzero Dale-constrained trainable recurrent excitation moves
the matched COBA/PING models beyond the current ~70% regime.

Use current main as the base. Build exp073 around the best surviving SHD
recipe from exp071/exp072, especially the cumulative-potential readout. Do
not rerun the old baseline unless needed for debugging; use exp071/exp072 as
the historical baseline.

Experiment:
- run matched COBA and PING cells with nonzero W_EE initialization and --
trainable-w-ee;
- keep W_in and W_out trainable as in the current training path;
- keep Dale signs enforced; do not use free signed recurrence;
- start with a modest nonzero W_EE initialization, e.g. --w-ee 0.03 0.01,
unless local inspection of current parameter scales suggests a safer nearby
value;
- keep input preprocessing, dataset split, readout, optimizer, batch size,
and evaluation protocol matched across COBA and PING;
- keep the comparison exploratory and single-seed unless I explicitly ask for
replication.

Protocol:
- implement through experiments/exp073.py and existing tools/snn CLI
capabilities if possible;
- do not edit tools/snn unless the current CLI cannot express nonzero
```

```
trainable W_EE cleanly; if a tools/snn change is required, stop and explain
exactly why before editing;
- run a local smoke/scout stage first with few epochs/samples to confirm
finite loss, active spiking, and W_EE actually changes during training;
- if both cells are finite and active, run a short RunPod exploratory stage,
fewer than 40 epochs for iteration speed;
- if accuracy looks meaningfully better than the exp071/exp072 historical
~70% regime without pathological firing or NaNs, run a 40-epoch confirmation
for the promising cell(s);
- COBA and PING may run in parallel on separate RunPod pods when useful;
- RunPod spending is authorized up to $40 total; reap all pods when finished.

Deliverables:
- artifacts/data/exp073 with provenance, numbers.json, reproducer, training
curves, firing-rate diagnostics, W_EE diagnostics, and matched rasters where
relevant;
- writings/exp073.typ as a cold-readable Demolab experiment report comparing
exp073 to the historical exp071/exp072 baseline;
- update the Spiking Heidelberg Digits collection so exp073 is the next
canonical experiment;
- build successfully;
- commit focused implementation/results changes, push the branch, and open a
new PR;
- do not merge unless I explicitly ask.

End at the review gate with:
- best/selected COBA and PING accuracies;
- whether trainable nonzero W_EE appears to move us beyond the ~70% regime;
- W_EE diagnostics showing it was nonzero and trained;
- firing-rate/pathology summary;
- exact compute spend;
- validation performed;
- PR link and rendered exp073 link.
```

Methods

The intended comparison inherited the exp071/exp072 SHD recipe: seed 42, 7340 development-training utterances, 816 held-out validation utterances, batch size 32, Adam learning rate 0.0004, 1 ms simulation steps, 1000 ms windows, matched input preprocessing, and the cumulative-potential readout with signed abstract readout weights and a trainable bias. The official SHD test set remained sealed and inaccessible to the runner.

The only intended new scientific ingredient was `--trainable-w-ee` with nonzero `--w-ee`. Dale's law stayed enabled; `W_EI`, `W_IE`, and `W_II` stayed frozen. COBA kept the no-inhibitory-loop recipe and no voltage-gradient dampening; PING kept the registered inhibitory loop and dampening 1000.

During implementation, the local CLI accepted `--w-ee` but training did not pass it into model construction. Commit `247e186` fixed that plumbing and added a focused regression test before this experiment ran.

Local gate

The final registered local scout used `--w-ee #r.config.w_ee_init.at(0) #r.config.w_ee_init.at(1)` and two smoke epochs on 128 training / 128 validation samples. The gate required both cells to be finite, active, non-saturated, and free of skipped or non-finite updates before any RunPod dispatch.

Cell	Passed	Selected acc.	E rate	I rate	Errors
COBA	`false`	7.03%	23.67 Hz	0 Hz	skipped/non-finite updates
PING	`true`	9.38%	3.95 Hz	16 Hz	none

COBA was active but failed the finite-update criterion. PING passed the local plumbing check. Because the mandate requires both cells to clear the smoke stage before cloud execution, no short RunPod stage was launched.

W_EE diagnostics

The saved smoke checkpoints confirm that `W_EE` was nonzero, Dale-constrained, and trainable enough to move from its reconstructed initialization:

Cell	Initial mean	Selected mean	Mean $|\Delta|$	Nonnegative
COBA	0.00000117	0.0002692	0.00026899	`true`
PING	0.00000117	0.00067511	0.00067479	`true`

The killed local scouts are archived under `artifacts/data/exp073/raw/killed_scouts/`. Their pattern was consistent: COBA failed the skipped-update criterion at every nonzero W_EE scale tried, while PING passed.

Scout	COBA passed	COBA selected	PING passed	PING selected
`0.3 0.01`	`false`	5.47%	`true`	8.59%
`0.03 0.01`	`false`	5.47%	`true`	8.59%
`0.003 0.001`	`false`	6.25%	`true`	9.38%
`0.0003 0.0001`	`false`	7.03%	`true`	9.38%

Conclusion

The requested matched experiment cannot honestly proceed to RunPod under the locked local-gate rule. Nonzero plastic recurrent excitation is not yet a safe plug-in addition to the current matched COBA/PING SHD recipe: COBA's no-inhibitory-loop cell develops pathological gradients before the pilot stage, while PING remains well behaved. Progress from here requires a new design authority decision, such as adding a stabilizer, changing COBA's recurrence recipe, or giving recurrent weights their own optimizer controls.

Paid compute spend: 0 USD. Active pods after this checkpoint: 0.